Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience building distributed cloud infrastructure, backend automation, and data pipelines using Java, Python, AWS, and Kubernetes. Experienced designing scalable, fault tolerant systems and building real-time ML/data pipelines with PySpark and FastAPI.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Distributed Systems & Streaming: PySpark, Apache Spark, Delta Lake, event driven architecture, REST APIs, gRPC.

Data & Storage: PostgreSQL, MySQL, Redis, schema validation (Zod), FastAPI.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube, Azure, Google Cloud Platform.

AI & Observability: Gemini, Vertex AI, PyTorch, scikit-learn, LangChain, CloudWatch, Datadog.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity, using a provider-agnostic abstraction across Gemini, GPT-4o, Llama, and Qwen.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services using CloudWatch and Datadog dashboards.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** with automated rollback and recovery.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PySpark, Delta Lake, FastAPI, Redis, Docker] Jul 2026

- Built distributed ML pipeline forecasting NYC taxi/rideshare demand across **263 zones** using a Spark-based medallion (bronze/silver/gold) architecture over a **2.3M-row** hourly dataset; served real-time predictions via FastAPI with Redis caching, cutting RMSE **43%** over baseline.

PolicyPilot – Policy Intelligence Portal [React, TS, Node, Gemini, MCP, Docker] May 2026

- Designed **17-tool MCP server** exposing structured LLM tool-calling over an evidence-grounded RAG pipeline, with Zod schema validation at every data boundary to eliminate runtime type-mismatch bugs across external tool calls.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).